DSA Tutorial - 2

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Topic: Binary Search

INPUT :

```
#include <stdio.h>

int binarySearch(int arr[], int low, int high, int target) {
    if (low > high) {
        return -1;
    }

    int mid = low + (high - low) / 2;

    if (arr[mid] == target) {
        return mid;
    }
    if (arr[mid] > target) {
        return binarySearch(arr, low, mid - 1, target);
    }
    return binarySearch(arr, mid + 1, high, target);
}

int main() {
    int n, target;
printf(“Viraj T B25ET1053”);
    printf("Enter the number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d sorted elements:\n", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
```

```
    printf("Enter the element to search: ");
    scanf("%d", &target);

    int result = binarySearch(arr, 0, n - 1, target);

    if (result != -1) {
        printf("Element found at index: %d\n", result);
    } else {
        printf("Element not found\n");
    }

    return 0;
}
```

OUTPUT :

